

Manejo do Sangramento Digestivo Baixo

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Prognóstico

- O sangramento cessa espontaneamente em 80 a 85% dos pacientes
- A taxa de mortalidade é de 2 a 4%



Causas de HDB

Causa	Porcentagem de Casos
Diverticulose	30–65
Colite isquêmica	5–20
Hemorroidas	5–20
Pólipos colorretais ou neoplasias	2–15
Angioectasias	5–10
Sangramento pós-polipectomia	2–7
Doença inflamatória intestinal	3–5
Colite infecciosa	2–5
Ulceração estercoral	0–5
Varizes colorretais	0–3
Proctopatia actínica	0–2
Colopatia induzida por AINEs	0–2
Lesão de Dieulafoy	Rara

n engl j med 376;11 nejm.org March 16,

2017



Formas de Apresentação

Hematêmese: sangramento através dos vômitos (ausente na HDB!)

Melena: sangue oxidado com escurecimento e fetidez

Hematoquezia: sangramento normalmente de pequena monta associado ao ato evacuatório

Enterorragia: hemorragia volumosa sem associação com o ato evacuatório

Sangramento oculto: detectado através de pesquisa laboratorial ou suspeitada a partir de anemia persistente

Fatores confundidores: bismuto, carvão, alcaçuz e ferro – fezes enegrecidas sem HDA



Formas Clínicas Conforme Etiologia

Anatômicas: (diverticulose)

Vascular: (angiodisplasia, isquêmica, induzida por radiação)

Inflamatório: (doença inflamatória intestinal, infecciosa)

Neoplásico



Estratificação de Risco - Escore de Oakland

Variável	Escore
Idade, anos	
< 40	0
40–69	1
> 70	2
Sexo	
• Feminino	0
• Masculino	1
Admissão prévia por HDB	
• Não	0
• Sim	1
Achados de exame retal digital	
• Sem sangue	0
• Com sangue	1
Frequência cardíaca, bpm	
• < 70	0
• 70–89	1
• 90–109	2
• > 110	3
Pressão arterial sistólica, mmHg	
• 50–89	5
• 90–119	4
• 120–129	3
• 130–159	2
• > 160	0
Hemoglobina, g/dL	
• 3.6–6.9	22
• 7.0–8.9	17
• 9.0–10.9	13
• 11.0–12.9	8
• 13.0–15.9	4
• ≥ 16.0	0



Escore SHA₂PE

Variável	Escore
Pressão sistólica < 100 mmHg	1
Hemoglobina (g/dL)	
< 10,5	2
10,5 – 12,0	1
Terapia antiplaquetária	1
Terapia anticoagulante	1
Pulso > 100 bpm	1
Sangramento no pronto-socorro	1



Manejo Geral - HDB

ABCD,
ressuscitação
volêmica,
estabilização



Exclusão de
fonte de HDA
exteriorizada
via baixa



Avaliação de
fonte baixa de
sangramento >
Colonoscopia



The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812

JANUARY 3, 2013

VOL. 368 NO. 1

Transfusion Strategies for Acute Upper Gastrointestinal Bleeding

Comparação randomizada de limiares de 7 g/dL versus 9 g/dL em 899 pacientes

- Mortalidade em 45 dias como desfecho primário.
- Exclusão de Pacientes com sangramento maciço e com vasculopatia periférica sintomática, acidente vascular cerebral ou ataque isquêmico transitório nos últimos 90 dias.

SUPERIORIDADE DO BRAÇO RESTRITIVO

- menos transfusão (49% vs 86%, $P < 0,001$)
- menor mortalidade (5% vs 9%, $P < 0,02$; [HR] 0,55, IC 95% 0,33–0,92),
- menos sangramento adicional (10% vs 16%, $P < 0,01$),
- menos reações transfusionais (3% vs 9%, $P < 0,001$)
- menos complicações cardíacas (11% vs 16%, $P < 0,04$)

N Engl J Med 2013;368:11-21

Restrictive versus liberal blood transfusion for gastrointestinal bleeding: a systematic review and meta-analysis of randomised controlled trials

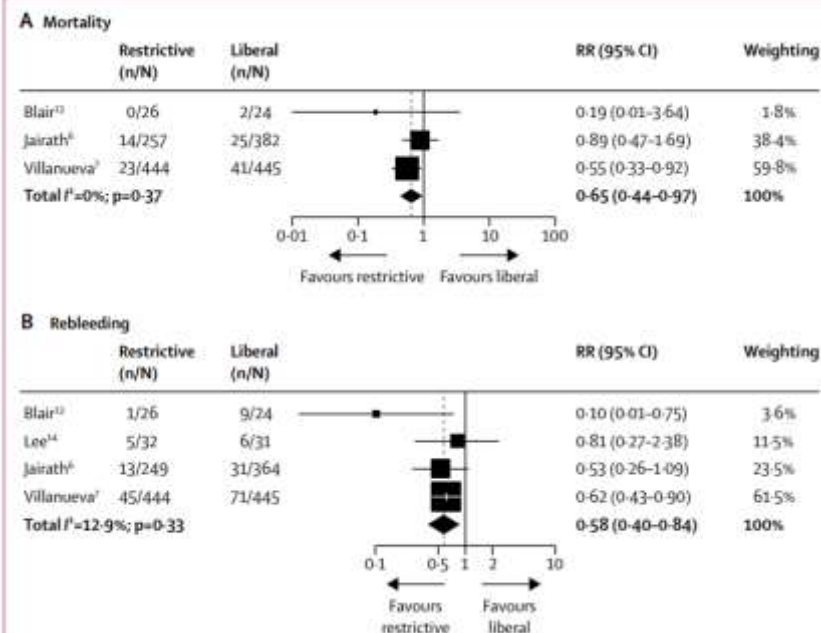
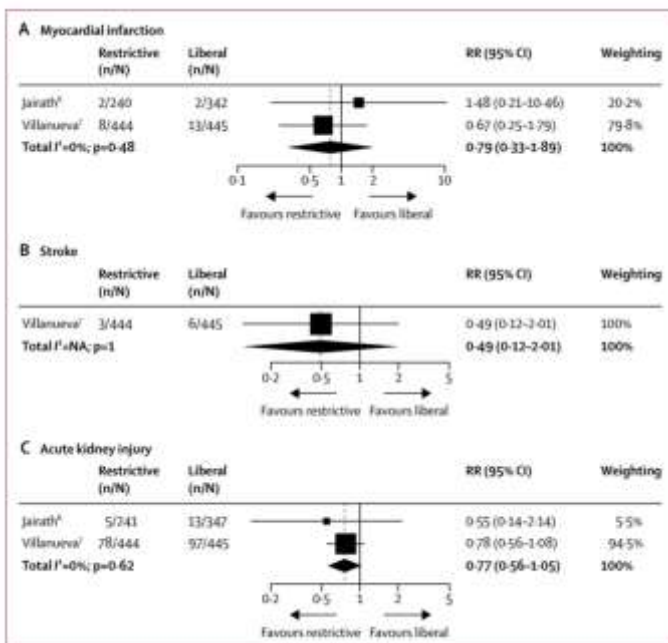
Ayodele Odutayo*, Michael J R Desborough*, Marialena Trivella, Adrian J Stanley, Carolyn Dorée, Gary S Collins, Sally Hopewell, Susan J Brunskill, Brennan C Kahan, Richard F A Logan, Alan N Barkun, Michael F Murphy, Vipul Jairath

	Country	Design	Intervention haemoglobin threshold	Follow-up	Outcomes reported
Blair et al, 1986 ¹³	UK	Single centre, parallel group	Restrictive: 80 g/L; liberal: no threshold (all received two units of red blood cells)	Not stated	Laboratory measures of coagulation (primary), mortality, rebleeding, red blood cell transfusions
Lee et al, 2014 ¹⁴	South Korea	Single centre, parallel group	Restrictive: 80 g/L; liberal: 100 g/L	Not stated	Rebleeding*
Jairath et al, 2015 ⁶	UK	Multicentre, cluster randomised	Restrictive: 80 g/L; liberal: 100 g/L	28 days	Feasibility (primary), mortality, rebleeding, acute myocardial infarction, stroke, transfusion reactions, acute kidney injury, bacterial infection, red blood cell transfusions, duration of hospital admission
Villanueva et al, 2013 ⁷	Spain	Single centre, parallel group	Restrictive: 70 g/L; liberal: 90 g/L	45 days	Mortality (primary), rebleeding, acute myocardial infarction, stroke, transfusion reactions, acute kidney injury, bacterial infection, red blood cell transfusions, duration of hospital admission
Villarejo et al, 1999 ¹⁵	Argentina	Single centre, parallel group	Restrictive: haematocrit 21%; liberal: haematocrit 28%	Not stated	Mortality, rebleeding, acute myocardial infarction, stroke, duration of hospital admission*

*No primary outcome specified for Lee et al¹⁴ or Villarejo et al.¹⁵

Restrictive versus liberal blood transfusion for gastrointestinal bleeding: a systematic review and meta-analysis of randomised controlled trials

Ayodele Odutayo*, Michael J R Desborough*, Marialena Trivella, Adrian J Stanley, Carolyn Dorée, Gary S Collins, Sally Hopewell, Susan J Brunskill, Brennan C Kahan, Richard F A Logan, Alan N Barkun, Michael F Murphy, Vipul Jairath



American College of Gastroenterology-Canadian Association of Gastroenterology Clinical Practice Guideline: Management of Anticoagulants and Antiplatelets During Acute Gastrointestinal Bleeding and the Periendoscopic Period

Neena S. Abraham, MD, MSc (Epi), FACP¹, Alan N. Barkun, MD, MSc (Epi), FACP, CAGP²,

Vitamin K antagonist reversal

1. For patients on warfarin who are hospitalized or under observation with acute GI bleeding, we suggest against FFP administration (conditional recommendation, very low certainty of evidence).
2. For patients on warfarin who are hospitalized or under observation with acute GIB, we could not reach a recommendation for or against PCC administration.
3. For patients on warfarin who are hospitalized or under observation with acute GIB, we suggest PCC administration compared with FFP administration (conditional recommendation, very low certainty of evidence).
4. For patients on warfarin who are hospitalized or under observation with acute GIB (upper and/or lower), we suggest against the use of vitamin K (conditional recommendation, very low certainty of evidence).



Blood transfusion: summary of NICE guidance

Smita Padhi *senior research fellow*¹, Sophia Kemmis-Betty *senior health economist*¹, Sharangini

Patients with thrombocytopenia who are bleeding

- Offer platelet transfusions to patients with thrombocytopenia who have clinically significant bleeding (World Health Organization bleeding grade 2—for example, prolonged epistaxis, extensive skin bleeding, haematemesis, or melaena) and a platelet count below $30 \times 10^9/L$. [*Based on the experience and opinion of the GDG.*]

Patients with thrombocytopenia who are having invasive procedures or surgery

- Consider prophylactic platelet transfusions to raise the platelet count above $50 \times 10^9/L$ in patients who are having invasive procedures or surgery. [*Based on very low to low quality evidence from RCTs and the experience and opinion of the GDG.*]



Impact of INR monitoring, reversal agent use, heparin bridging, and anticoagulant interruption on rebleeding and thromboembolism in acute gastrointestinal bleeding

Naoyoshi Nagata , Toshiyuki Sakurai, Shiori Moriyasu, Takuro Shimbo, Hidetaka Okubo, Kazuhiro Watanabe, Chizu Yokoi, Mikio Yanase, Junichi Akiyama, Naomi Uemura

	Propensity score for each treatment	Rebleeding risk		Thromboembolic risk	
	C-statistic (95%CI)	Adjusted odds ratio (95%CI)	P	Adjusted odds ratio (95%CI)	P
DOAC use	0.81 (0.74–0.88)	0.8 (0.2–2.5)	0.648	0.6 (0.06–5.9)	0.646
PT-INR (onset) ≥ 2.5	0.74 (0.65–0.83)	0.7 (0.2–2.3)	0.523	7.3 (1.5–35.3)	0.013
Reversal agent (vitamin K antagonist) use	0.92 (0.87–0.97)	1.4 (0.3–7.3)	0.722	1.2 (0.2–9.1)	0.840
Heparin bridging	0.77 (0.68–0.85)	0.8 (0.3–2.3)	0.648	1.3 (0.3–5.8)	0.767
Anticoagulant interruption before endoscopy	0.79 (0.71–0.86)	3.3 (0.7–16.1)	0.081	NA [†]	NA [†]



RESEARCH ARTICLE

Originally Published 18 March 2009 |

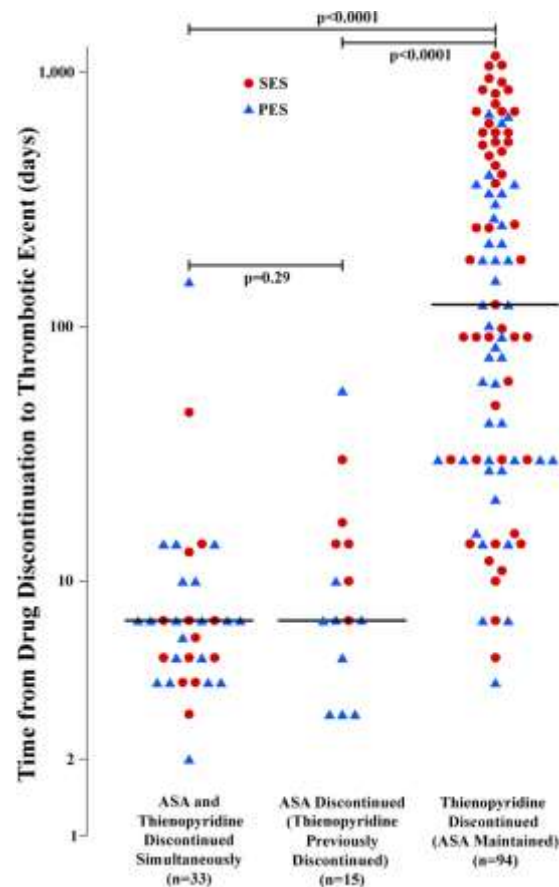
Check for updates

Safety of Short-Term Discontinuation of Antiplatelet Therapy in Patients With Drug-Eluting Stents

Mark J. Eisenberg, MD, MPH, Pierre R. Richard, BSc, Danielle Liberman, PhD, and Kristian E. Fikri, MSc

AUTHOR INFO & CITATIONS

Circulation • Volume 119, Number 12 • <https://doi.org/10.1161/CIRCULATIONAHA.108.812067>

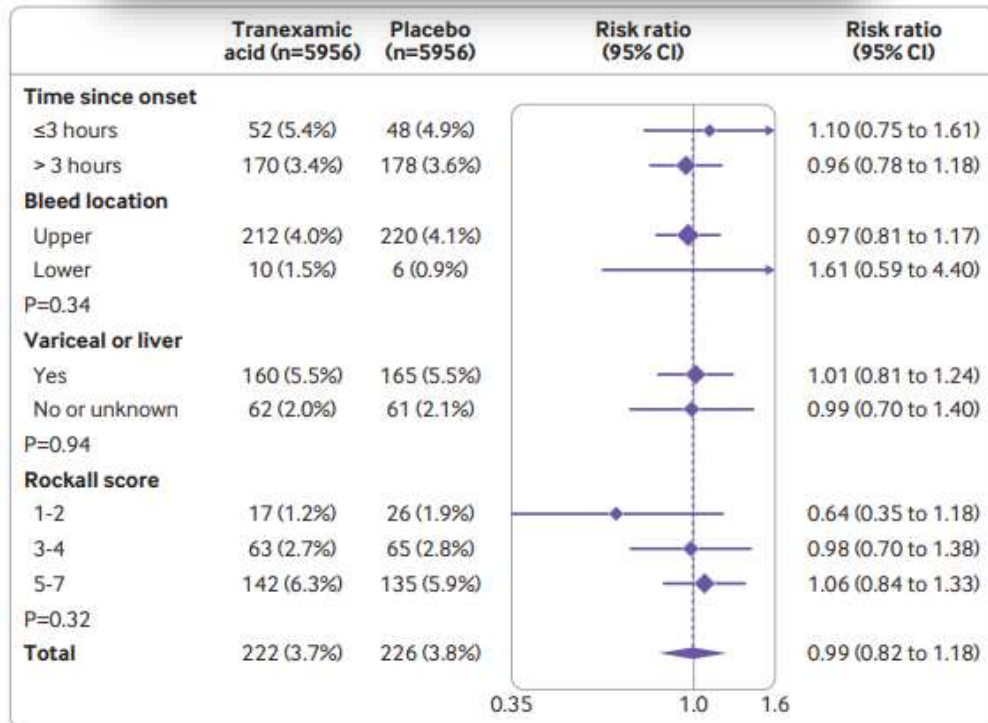


Manejo de antiagregantes

- Aspirina profilaxia primária = suspender
- Aspirina cardíaca secundária = continuar durante hospitalização, descontinuar em casos graves temporariamente
- Risco cardiovascular / trombótico = discussão multidisciplinar
- Terapia dupla: preferência por continuar aspirina enquanto for suspenso antagonista P2Y12
 - > Retornar em no máximo 5 dias se Stent no último ano

Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial

The HALT-IT Trial Collaborators*



Lancet 2020; 395: 1927–36



ORIGINAL CONTRIBUTIONS: COLON/SMALL BOWEL

Randomized Trial of Urgent vs. Elective Colonoscopy in Patients Hospitalized With Lower GI Bleeding

Laine, Loren MD¹; Shah, Abbid MD¹

Eighty-five eligible patients had urgent upper endoscopy; 13 (15%) had an upper source



Manejo Geral HDB

Descartar HDA previamente se apropriado:

*Risk factors of UGI bleeding include history of peptic ulcer bleeding, portal hypertension, BUN/Cr>30, aspirin/NSAID use, and positive nasogastric lavage



Does This Patient Have a Severe Upper Gastrointestinal Bleed?

F. Douglas Srygley, MD

Charles J. Gerardo, MD

Tony Tran, MD

Deborah A. Fisher, MD, MHS

Modelo predição para HDA

Laboratory findings

Serum urea nitrogen:
creatinine ratio
>30^{8,29-33,b}

51 (26 to 75)

93 (87 to 99)

7.5 (2.8-12.0)

0.53 (0.28-0.78)

Menor probabilidade de HDA

Clots in stool⁸

15 (14-15)

99.2 (96.0-99.9)

0.05 (0.01-0.38)

1.2 (1.1-1.2)



REVISIÓN

Eficacia diagnóstica de la sonda nasogástrica y de la relación nitrógeno ureico en sangre (BUN)/creatinina para distinguir el origen alto o bajo de la hemorragia digestiva

Salvador Machlab^{1,2}, Pilar García-Iglesias^{1,2}, Eva Martínez-Bauer^{1,3}, Rafael Campo^{1,3}, Xavier Calvet^{1,3}, Enric Brullet^{1,2}

Tabla 3. Resultado de la eficacia diagnóstica en los estudios seleccionados

Estudio (año)	Test	Sensibilidad % (IC 95%)	Especificidad % (IC 95%)	Valor predictivo positivo % (IC 95%)	Valor- predictivo negativo % (IC 95%)	Razón de verosimilitud negativa % (IC 95%)	Razón de verosimilitud positiva % (IC 95%)
Richards <i>et al.</i> ²⁰ (1990)	BUN/Cr	37,8 (27,6-49,2)	100 (93,1-100)	100 (87,9-100)	53,1 (43,3-62,6)	0,63 (0,52-0,74)	NC
Witting <i>et al.</i> ²¹ (2006)	BUN/Cr	39,1 (31-47,7)	93,9 (89,7-96,5)	80,6 (69,1-88,6)	70,3 (64,6-75,5)	0,65 (0,56-0,76)	6,41 (3,56-11,56)
	SNG	42 (32-51)	91 (83-95)	81	61	0,56 (0,46-0,68)	11,1 (4,1-30)
Aljebreen <i>et al.</i> ¹⁸ (2004)	SNG	68	54	41	78	0,61	1,44
Kessel <i>et al.</i> ¹⁹ (2016)	SNG	27,8 (23,6-32,5)	86	99	2	0,84	2



**Hematoquezia
Grave**

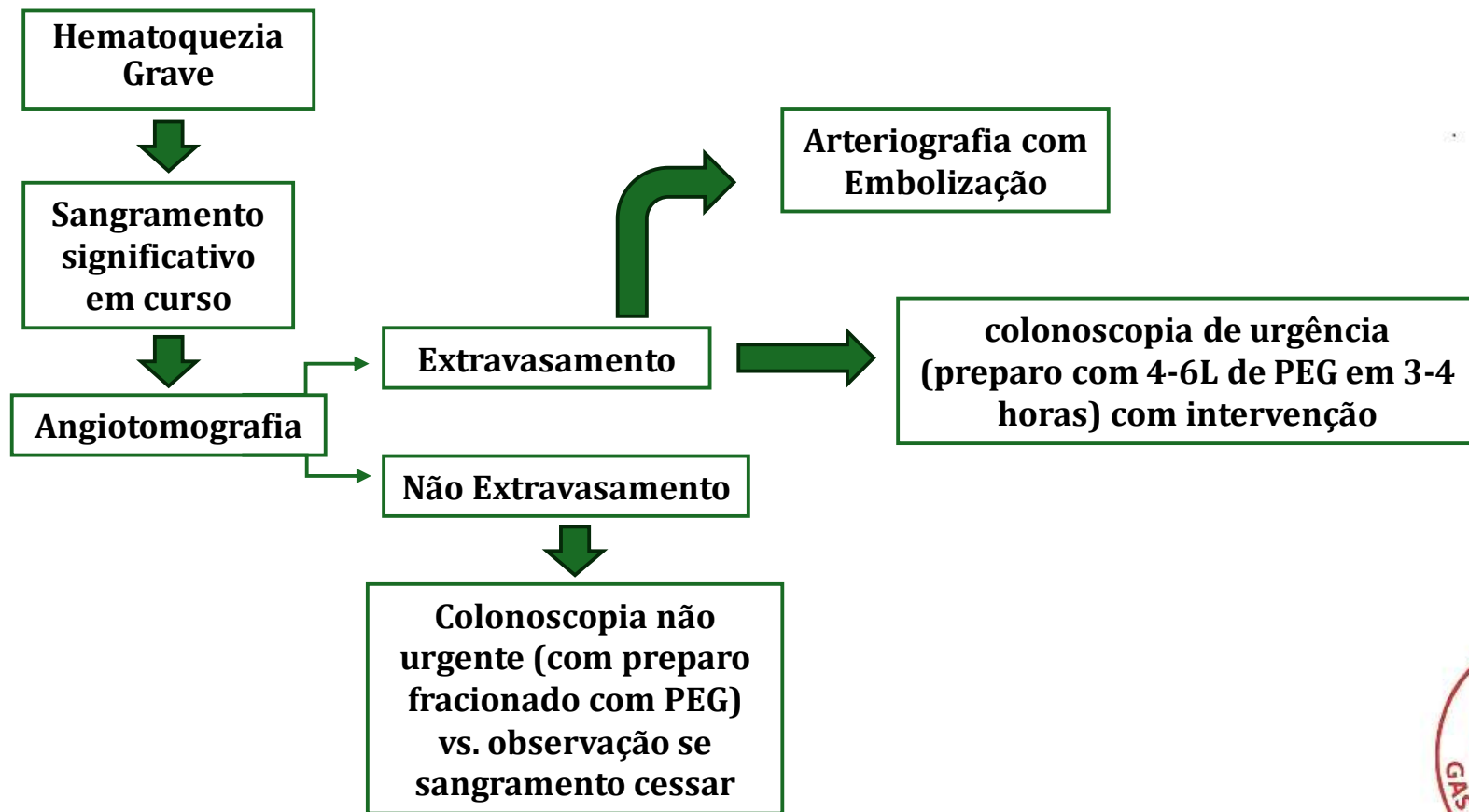


**Sangramento
leve ou
interrompido**



**Colonoscopia não
urgente (com preparo
fracionado com PEG)
vs. observação se
sangramento cessar**





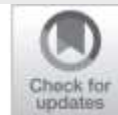
Identifying Bleeding Etiologies by Endoscopy Affected Outcomes in 10,342 Cases With Hematochezia: CODE BLUE-J Study

Nagata, Naoyoshi MD, PhD^{1,2}; Kobayashi, Katsumasa MD³; Yamauchi, Atsushi MD⁴; Yamada, Atsuo MD, PhD⁵; Omori

Initial colonoscopic examination	9,066 (87.7)	10,342	0
Time to first colonoscopy (hr)	16 (4–32)	9,066	0
SRH on initial endoscopy	2,801 (30.9)	9,066	0
SRH, active bleeding	1,489 (16.4)	9,066	0
SRH, visible vessel	535 (5.9)	9,066	0
SRH, adherent clot	829 (9.1)	9,066	0



Early Colonoscopy for Diverticular Bleeding Does Not Reduce Risk of Postdischarge Recurrent Bleeding: A Propensity Score Matching Analysis



Neha Nigam,^{*} Sandra A. Ham,[‡] and Neil Sengupta^{*}

	Total cohort (n = 16,640)	Early colonoscopy (n = 8320)	Delayed colonoscopy (n = 8320)	P value
Procedures, n (%)				
Colonoscopy with intervention	916 (6)	264 (3)	652 (8)	<.0001

ORIGINAL ARTICLE

Outcomes of Early Versus Delayed Colonoscopy in Lower Gastrointestinal Bleeding Using a Hospital Administrative Database

Neha Nigam, MD,^{*} Parita Patel, MD,[†] and Neil Sengupta, MD^{*}

Outcomes				
30-day mortality [n (%)]	31 (3%)	6 (2%)	25 (3%)	0.67
Colonoscopy with intervention [n (%)]	40 (3%)	18 (6%)	22 (2%)	0.004



Effectiveness of risk scoring systems in predicting endoscopic treatment in colonic diverticular bleeding

Noriaki Oguri,^{*,†} Takashi Ikeya,^{*} Daiki Kobayashi,[†] Kazuki Yamamoto,^{*} Takaaki Yoshimoto,^{*} Ayaka Takasu,^{*} Takeshi Okamoto,^{*} Yasutoshi Shiratori,^{*} Shuhei Okuyama,^{*} Koichi Takagi,^{*} Kenji Nakamura[§] and Katsuyuki Fukuda^{*}

Table 1 NOBLADS score and Strate score

NOBLADS score (all factors are 1 point)

No abdominal pain	No diarrhea (< 3 times)
NSAIDs	Systolic blood pressure ≤ 100 mmHg
Albumin < 3 g/dL	Antiplatelet agents (nonaspirin)
Charlson comorbidity index ≥ 2	Syncope

Strate score (all factors are 1 point)

Heart rate ≥ 100 /min	Systolic blood pressure ≤ 115 mmHg
Syncope	No abdominal pain
Aspirin	Charlson comorbidity index ≥ 3
Hemorrhage more than 4 h	

NSAIDs, non-steroidal anti-inflammatory drugs.



Effectiveness of risk scoring systems in predicting endoscopic treatment in colonic diverticular bleeding

Noriaki Oguri,^{*,†}  Takashi Ikeya,^{*}  Daiki Kobayashi,[†] Kazuki Yamamoto,^{*} Takaaki Yoshimoto,^{*} Ayaka Takasu,^{*} Takeshi Okamoto,^{*} Yasutoshi Shiratori,^{*} Shuhei Okuyama,^{*} Koichi Takagi,^{*} Kenji Nakamura[§] and Katsuyuki Fukuda^{*}

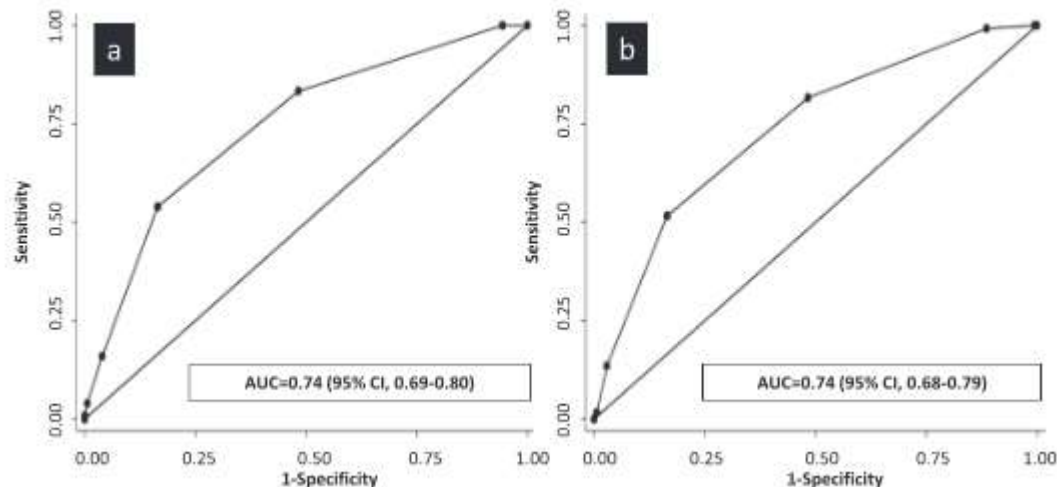


Table 3 The ABC score for prediction of 30-day mortality

Variable	Assigned score
Age	
60–74 years	1
≥75 years	2
Blood tests	
Urea >10 mmol/L	1
Albumin <30 g/L	2
Creatinine	
100–150 µmol/L	1
>150 µmol/L	2
Comorbidity	
Altered mental status	2
Liver cirrhosis	2
Disseminated malignancy	4
ASA score	
3	1
≥4	3

ABC, age, blood tests and comorbidities; ASA, American Society of Anaesthesiologists.

Gut 2021;**70**:707-716



Evidências para emprego de colonoscopia precoce

Table 4. Meta-analyses comparing urgent (<24 hours) to elective (>24 hours) colonoscopy in LGIB

Study	No. in each arm (U vs E)	Diagnostic yield ^a	Rebleeding	LOS	PRBC	Endoscopic intervention	Mortality
Analysis limited to RCTs							
Kherad et al.	230/236	ND	ND	ND	ND	ND	ND
Anvari et al.	228/235	ND	ND	ND	ND	ND	ND
Tsay et al.	228/235	ND	ND			ND	ND
Combined analysis of observational studies and RCTs							
Anvari et al.	63,105/66,170	+U		+U	ND	ND	+U
Roshan Afshar et al.	9,889/14,630	+U	ND	+U	ND	+U	ND
Seth et al.	9,498/13,921	ND	ND	ND		ND	ND
Kouanda et al.	10,172/14,224	ND	ND	ND	ND	+U	ND
Sengupta et al.	422/479	+U	ND	ND	ND	+U	ND

E, elective; LGIB, lower gastrointestinal bleeding; LOS, length of stay; ND, no significant difference between groups; PRBC, packed red blood cell transfusion; RCT, randomized controlled trial; U, urgent.

Comments: +U indicates that the results favored urgent colonoscopy; ND indicates that there was no significant difference seen between groups.

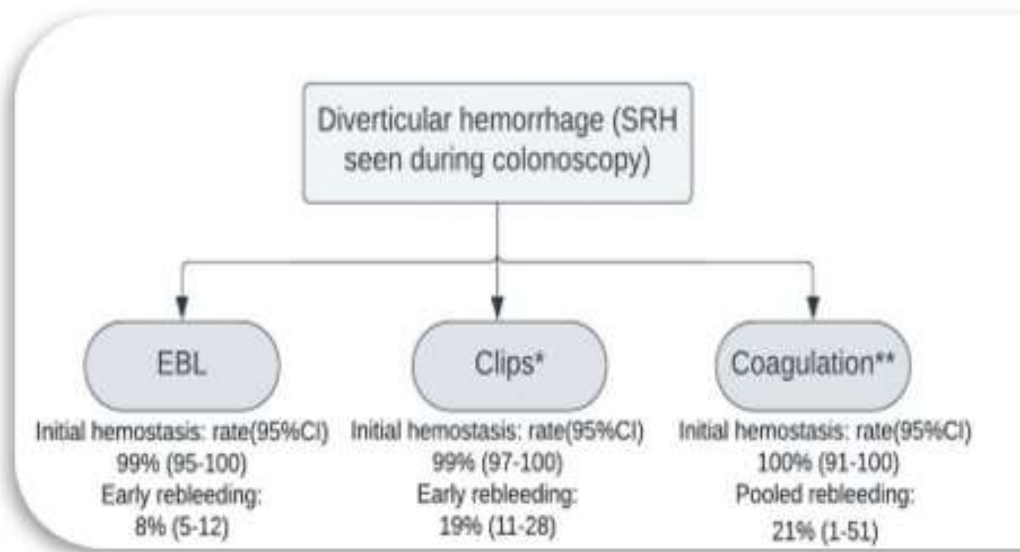
^aDiagnostic yield defined as definite or probable cause of acute LGIB.

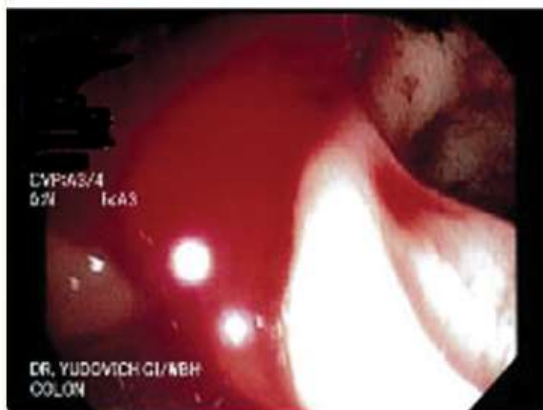
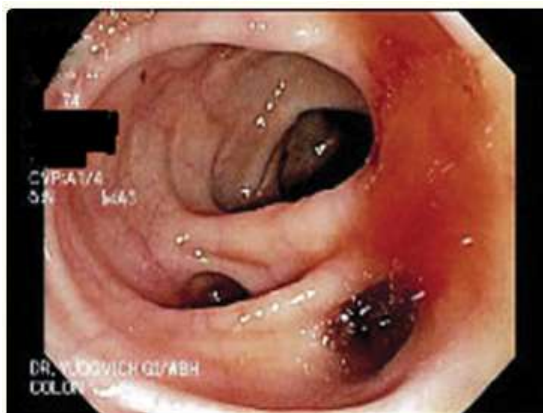
Preparo Colônico

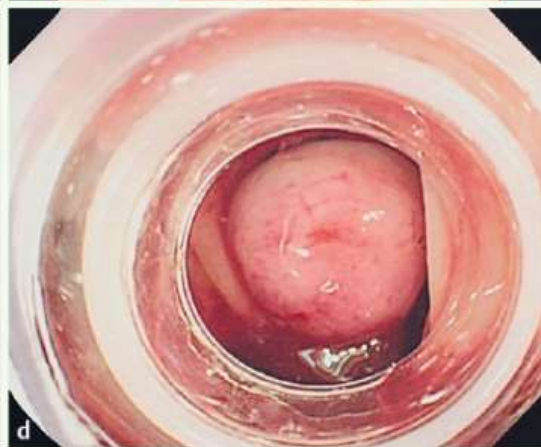
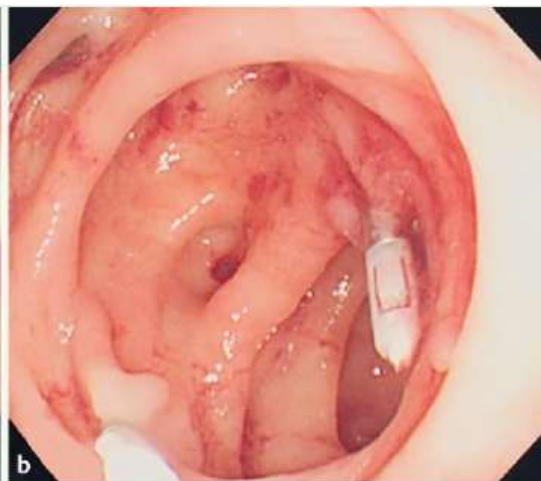
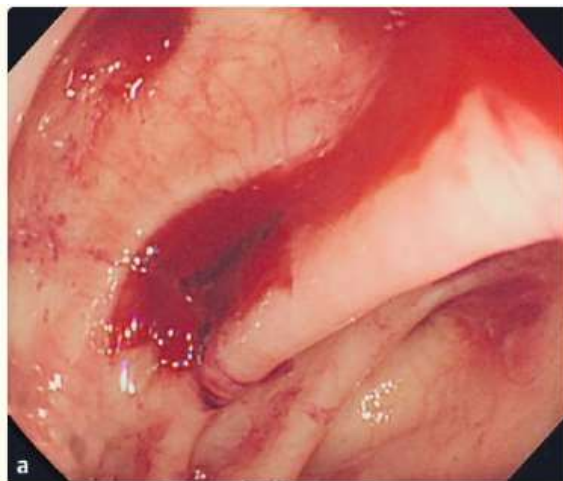
- Purgativos: bisacodil (12 horas antes do exame, efeito 6-8 horas)
- Osmóticos /passagem: PEG, lactulona, Manitol (1:1)
- Retrógrado – enterocisma: pacientes debilitados, ocluídos, sem via oral



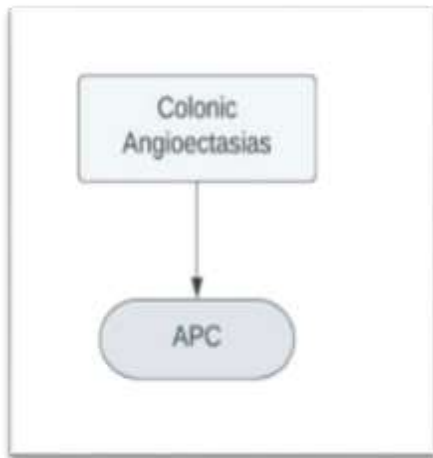
Manejo Específico por Lesão - HDB



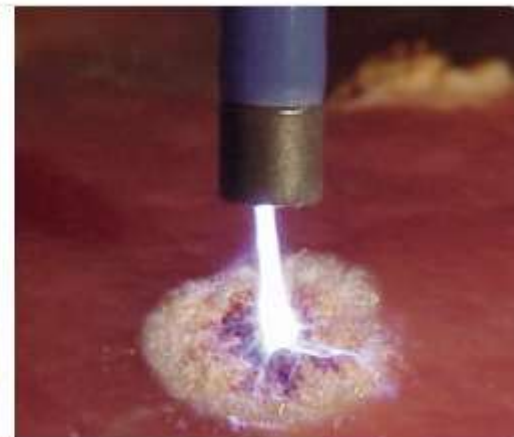
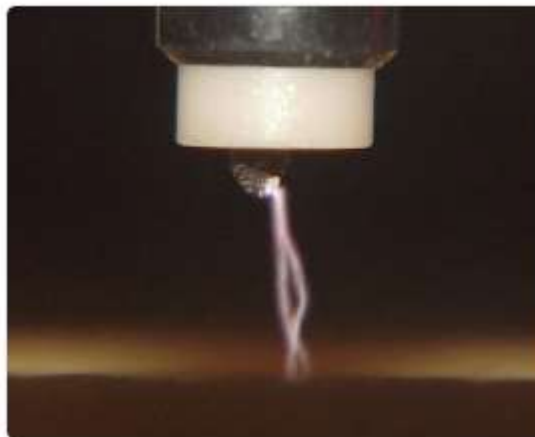




Manejo Específico p



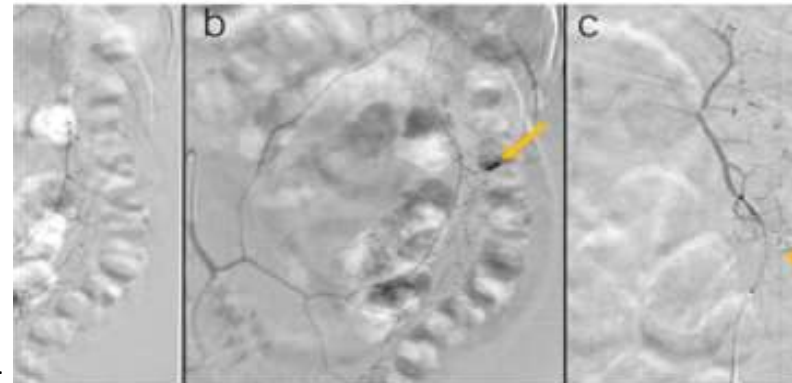
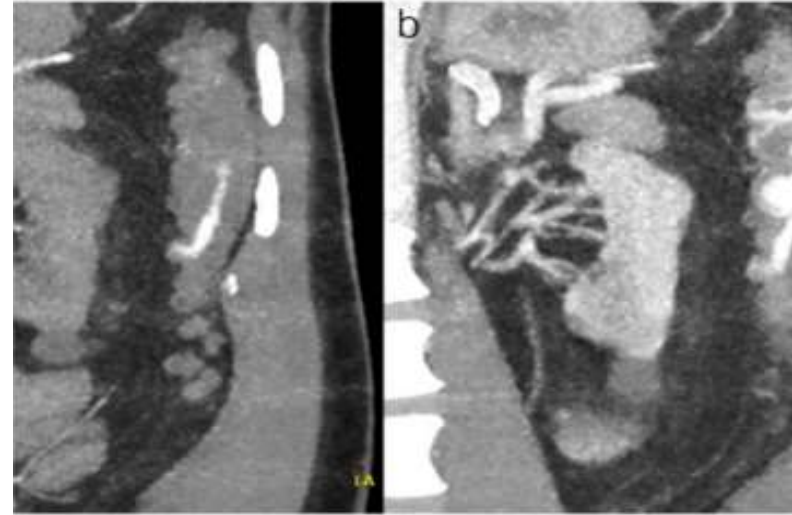
Am J Gastroenterol 2023;118:208–231



Angiografia: requer perda de sangue ativa de 0,5 a 1,0 mL/minuto

- Eficácia variável com o tempo de arteriografia após TC positiva
- Sensibilidade cai pela metade em sangramentos intermitentes
- Finalidade diagnóstica e terapêutica

**RISCO DE TROMBOEMBOLISMO, LESÃO VASCULAR,
ISQUEMIA E PERFURAÇÃO INTESTINAL**



Cintilografia de glóbulos vermelhos: detecta sangramento que ocorre a uma taxa de 0,1 a 0,5 mL/minuto com Tn99m

- Exame dura 24 horas com múltiplas capturas

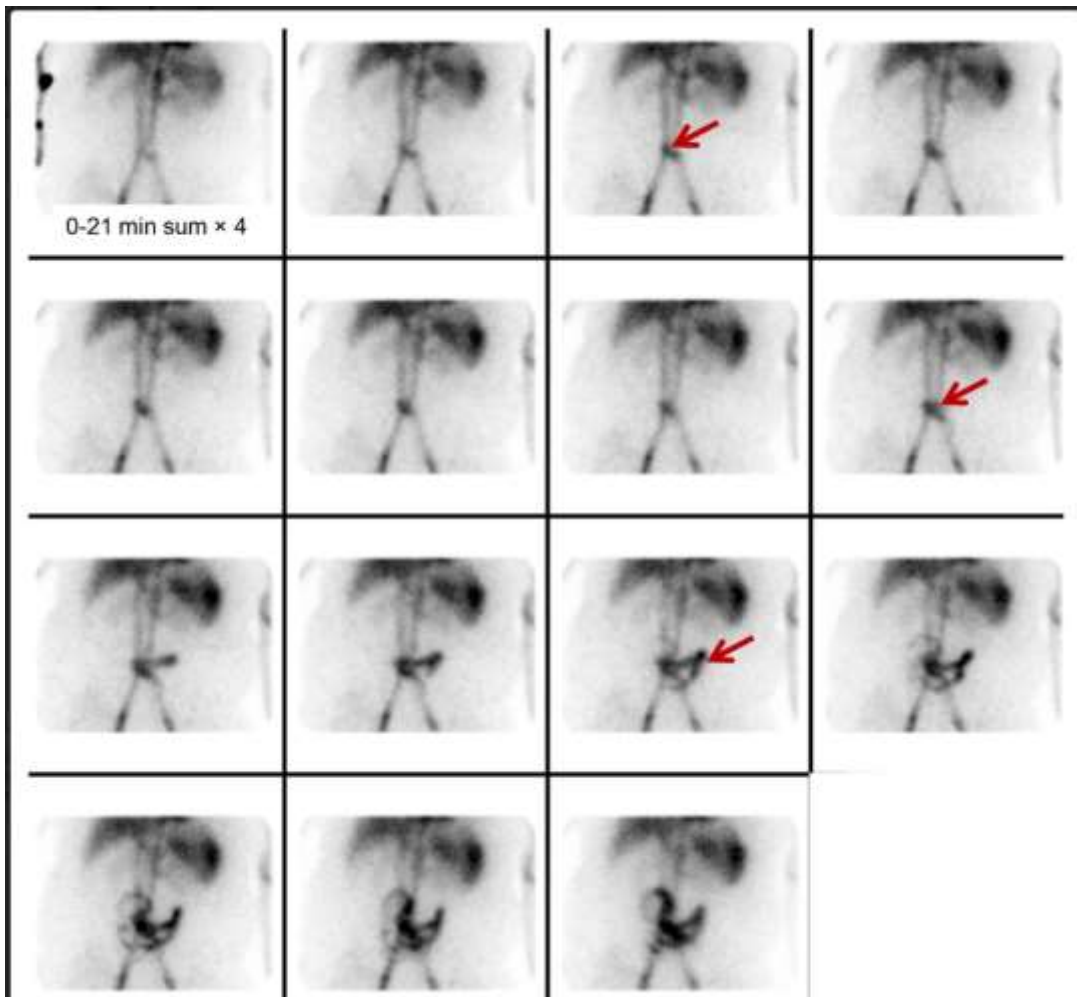
- Não necessita de preparo colônico

- Alta sensibilidade, porém encontra a “região anatômica” e não o ponto exato

➤ Arteriografia deve ser realizada imediatamente!!

**PODE ENCONTRAR SANGRAMENTO EM
ATÉ 80% DOS CASOS, PORÉM NÃO É
TERAPÊUTICO**





Sangramento Indeterminado

Repetição de EDA e colonoscopia -> se exames incompletos ou inadequados

Enteroscopia por pressão com visualização até 80 cm de jejuno

Estabilidade hemodinâmica: Enteroscopia por cápsula de vídeo sem fio

Instabilidade hemodinâmica: angiografia? Tratamento cirúrgico?



Falha em Colono e Radiologia Intervencionista

- Colectomia segmentar (85%) para os casos em que a laparotomia/arteriografia localizou o foco
- Colectomia subtotal (15%) em casos refratários sem identificação de foco
- Cirurgia também pode ser necessária para tratar complicações da arteriografia!

J Am Coll Radiol 2021;18:S139-S152



Obrigado!

Dúvidas?

Encaminhar para
gastrointensiva@gmail.com

